

Aakash Patil

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EDUCATION

Masters of Science in Computer Science

Aug 2025 – Present

George Mason University, Fairfax VA

GPA: 4.0

- **Relevant Coursework:** Analysis of Algorithms, Machine Learning, Deep Learning, Artificial Intelligence

Bachelors in Computer Engineering (Minor in AI&ML)

Aug 2021 – May 2025

Xavier Institute of Engineering, Mumbai

GPA: 3.2

- **Relevant Coursework:** Data Structures, Database Management Systems, Statistics & Probability, Big Data Analysis, Machine Learning, Operating Systems, Computer Networks

TECHNICAL SKILLS

Languages: Python, SQL, C++

Analytics & Visualization: Tableau, Power BI, Matplotlib, Seaborn

Data Tools: Pandas, NumPy, SciPy, Scikit-learn, Excel

Cloud Platforms: AWS, Azure, Google Cloud Platform

Databases: MySQL, MongoDB, DynamoDB, AWS S3, NoSQL

Platforms & Other: Docker, Jenkins, FastAPI, Streamlit, AWS Lambda, Git, ETL Pipelines

EXPERIENCE

Data Analytics Intern

June 2023 – July 2023

IBM SkillsBuild

Mumbai

- Conducted exploratory data analysis on image datasets using Pandas and Matplotlib, identifying quality metrics that guided preprocessing decisions
- Built data visualization dashboards to track model training performance, communicating insights to cross-functional team members
- Cleaned and transformed 10K+ image records through standardized ETL workflows, improving data consistency for downstream pipelines

PROJECTS

StyleSnap – AI Outfit Detection & Recommendation | Python, Polars, FastAPI, Gemini API

Feb 2025

- Processed and structured fashion image metadata using Polars and NumPy, building data pipelines to feed a YOLOv8 detection model
- Integrated Google Gemini API to generate context-aware outfit recommendations, transforming raw detection outputs into user-facing insights
- Deployed a FastAPI-based application enabling users to upload images and receive ranked recommendations, tracking usage metrics via logging

PPXAI – Privacy-Preserving Explainable AI | Python, Pandas, SHAP, LIME

July 2024 – May 2025

- Performed statistical analysis on healthcare prediction models, evaluating accuracy, precision, and recall across patient subgroups
- Generated SHAP-based feature importance reports to identify the top contributing variables in clinical prediction models, enabling stakeholder review
- Documented analytical findings on privacy-accuracy trade-offs, producing reproducible analysis notebooks for the research team

ERS – Content-Based Recommendation System | Python, NumPy, Pandas, Scikit-learn

Jan 2023 – May 2023

- Engineered TF-IDF feature vectors from unstructured text data and computed cosine similarity matrices to power a recommendation engine
- Performed A/B-style evaluation of recommendation quality metrics, iterating on feature selection to improve user satisfaction

Image Super Resolution Analytics | Python, Pandas, Matplotlib, TensorFlow

July 2023 – Nov 2023

- Tracked and visualized GAN training metrics (PSNR, SSIM, loss curves) across 50+ training epochs using Matplotlib dashboards
- Built a Streamlit reporting tool for non-technical users to compare input vs. output image quality side-by-side